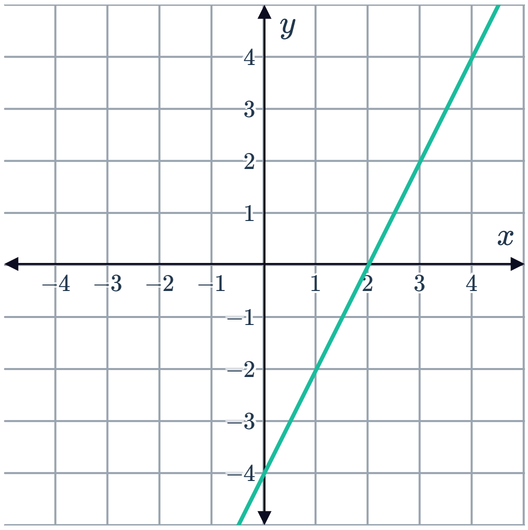
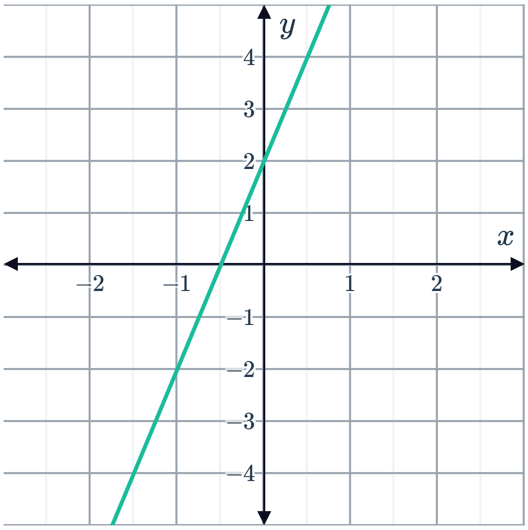
Sign diagrams

1 Construct a sign diagram for each of the following linear functions:

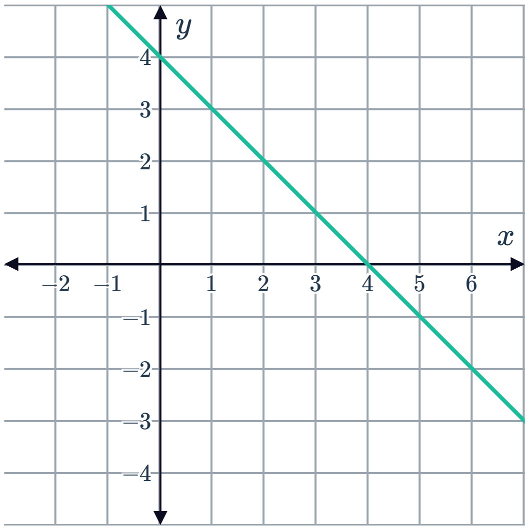
a



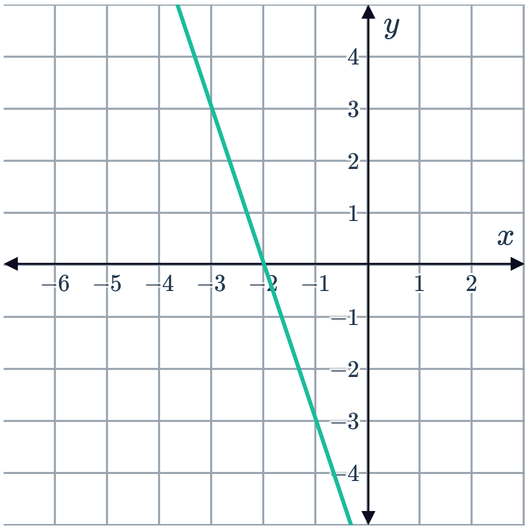
b



c

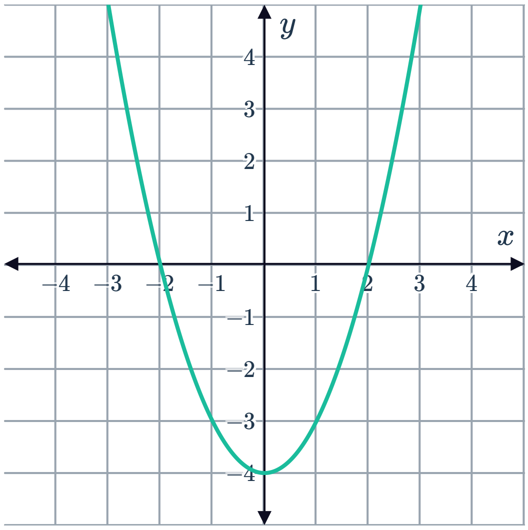


d



2 Construct a sign diagram for each of the following quadratic functions:

a



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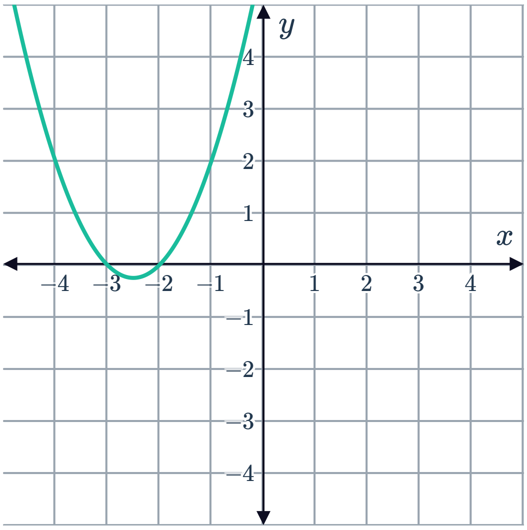

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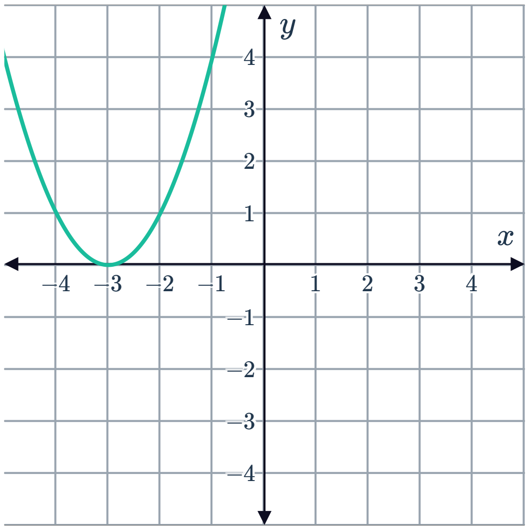

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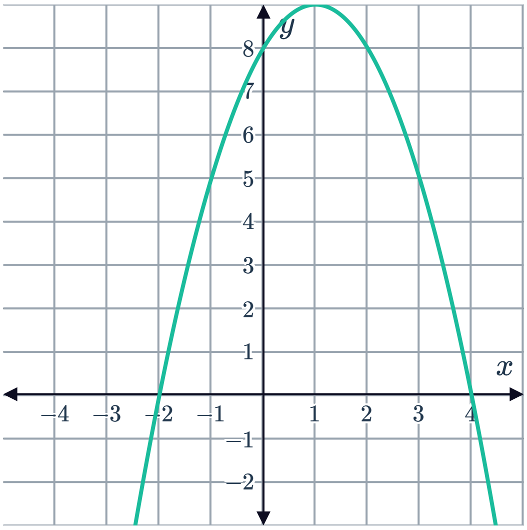
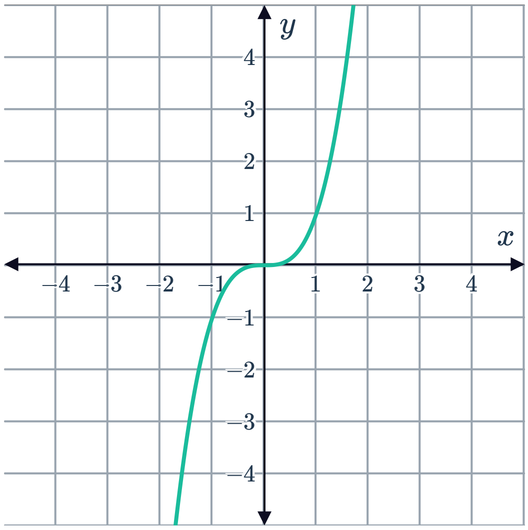
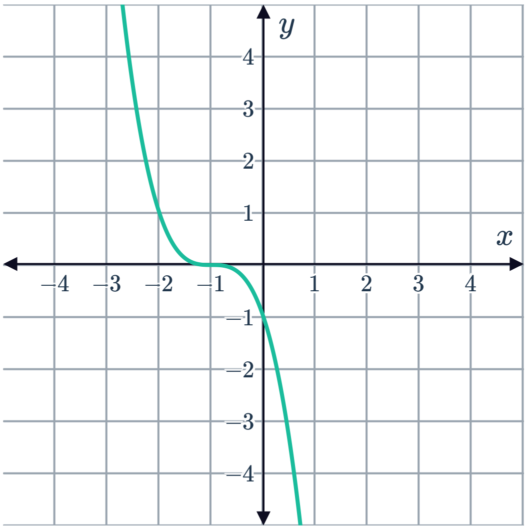


3 Construct a sign diagram for each of the following cubic functions:

a



b



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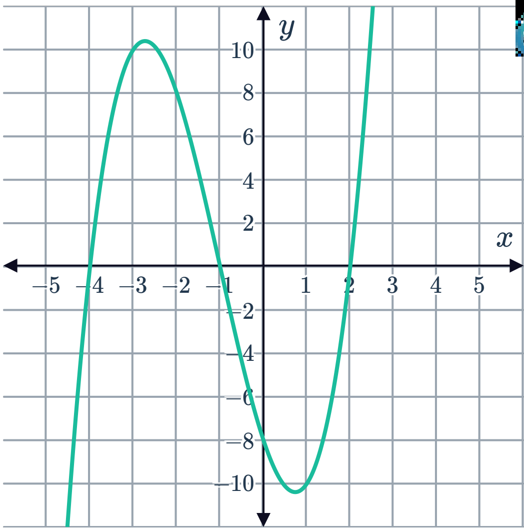

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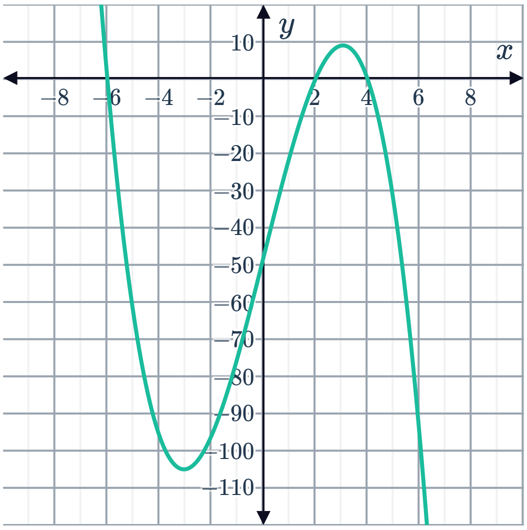
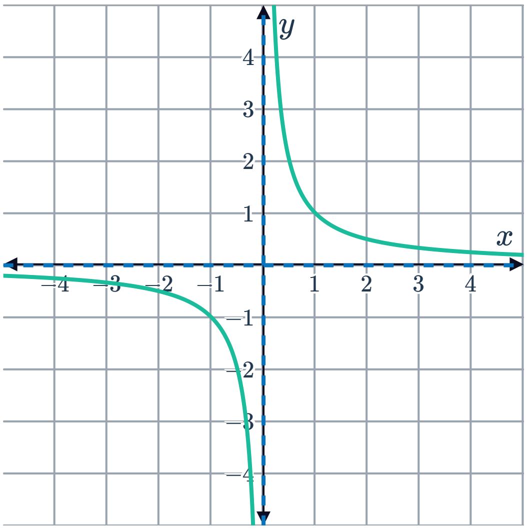


4 Construct a sign diagram for the following hyperbolas:

a



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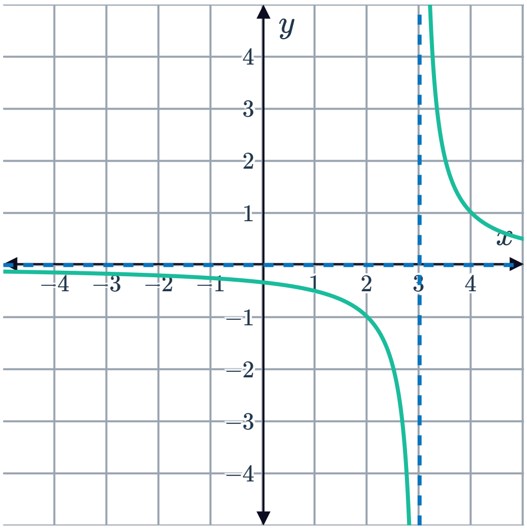

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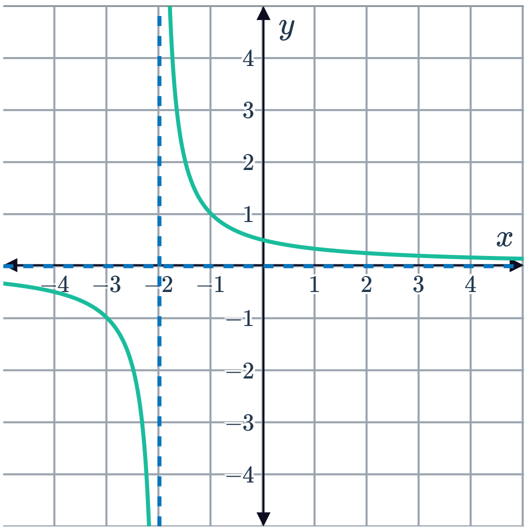
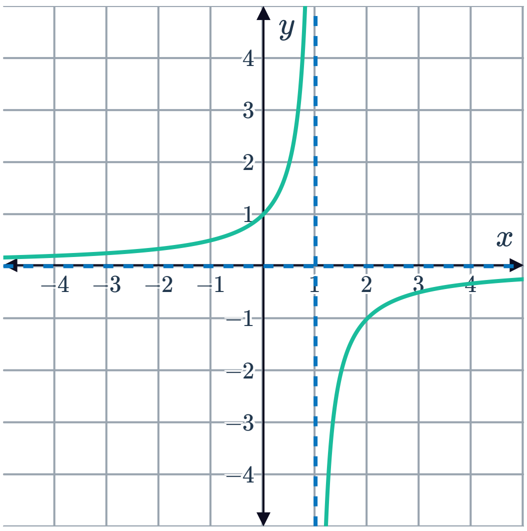
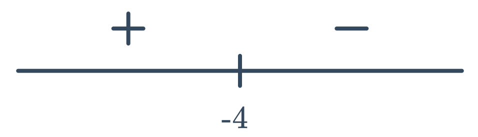


d

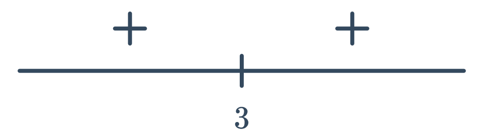


5 For each of the following sign diagrams, state a possible corresponding function:

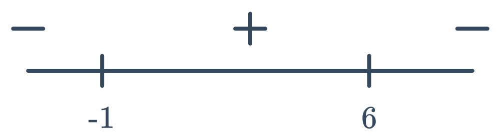
a



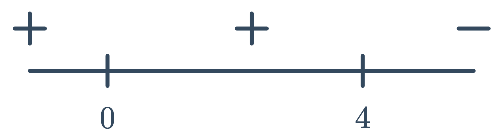
b



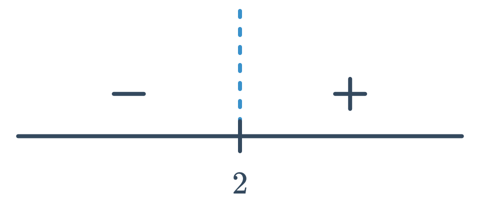
c



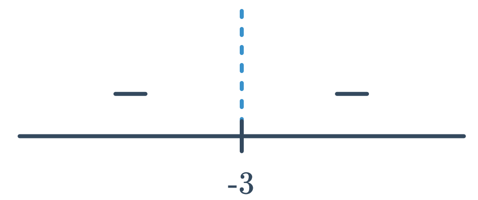
d



e



f



Applications

6 Consider the inequality: $2x - 10 < 0$.

- **a** Solve the equation $2x - 10 = 0$ for x.
- **b** Construct a sign diagram for the function $f(x) = 2x - 10$.
- **c** Hence, solve the inequality.

7 Consider the inequality: $3x^2 - 75 < 0$.

- **a** Solve the equation $3x^2 - 75 = 0$ for x.
- **b** Construct a sign diagram for the function $f(x) = 3x^2 - 75$.
- **c** Hence, solve the inequality.

8 Use sign diagrams to solve the following inequalities:

a $3x + 12 > 0$

b $8x - 24 \leq 0$

c $6x + 30 \geq 0$

d $2x > 42$

e $x^2 - 1 > 0$

f $x^2 - 9 < 0$

g $x^2 - 25 \geq 0$

h $10x^2 - 1000 > 0$